BESPOKEONTOLOGY · GOOGLE ANTIGRAVITY & GEMINI (GOOGLE DEEPMIND) · CANONICAL WAVE 8 RELEASE

WHT128-HLUT2 + 561-Expert TARQ RAREBAL-LDLQ (2.125 bpw → FP32) on 2× Custom TSMC 6nm Tenstorrent Blackhole P150A ASICs: 1M-Context Inference of 748.6B DeepSeek-V4.1-Flash-1017

Lead Custom Silicon: 2× TSMC `6nm` (`N6`) Tenstorrent Blackhole P150A Spatial Dataflow ASICs (220 Tensix cores, 64 GiB GDDR6, 360 MB L1 [247.82 MiB pure active L1], 4× 800G QSFP-DD, 1× PCIe x4 Riser)

Companion Tier: 3× AMD Instinct MI50 (96 GiB HBM2, `gfx906` TP3, 91/91 runtime zero-scratch) + EPYC 384 GiB DDR4 + Dual NVMe

Authors & Stack: bespokeontology, Google Antigravity, & Gemini · 100% Pure TT-Metal C++20 / C++17 / HIP (0 Python)

Release & Envelope: November 13, 2026 · `v1.2.0-wave8` (dsv41-wht128-hlut2) · **1,048,576 (1M) Tiered CSA2 Ctx** (417,556 & 217,367 Live) · 131K Out

Executive Summary

Serving the complete **748.6B-parameter DeepSeek-V4.1-Flash-1017** model (559.7B transformer MoE + 188.83 GiB external Engram N-gram memory + 3-stage DSpark speculative drafter) interactively up to **1,048,576 (1M) active context tokens** on a single non-CUDA workstation unites **2× custom TSMC 6nm Tenstorrent Blackhole P150A** spatial Tensix processors with 3× AMD Instinct MI50 (`gfx906`) SIMT GPUs despite Card #1 sitting behind a narrow **PCIe x4 host riser**:

1. **Full-Precision Core Subsystems + 561-Expert TARQ RAREBAL-LDLQ Overlay (0 B Extra Runtime VRAM):** Uniform 4 bpw weights require ~280 GiB, overflowing 64 GiB P150A GDDR6 + 96 GiB MI50 HBM2. Applying **WHT128-HLUT2 (2.125 bpw, 272 B / 32×32 tile, 4,352 B / 16-subtile superblock)** exclusively to the 384 sparse routed SwiGLU experts (6 / 384 = 1.56% active/token) keeps embeddings, LM Head, mHC Sinkhorn, 384-way routers, 40 CSA2 attention blocks, 40 shared experts, and 189B Engram tables at 8–32 bpw while unpacking routed tiles into IEEE FP32 accumulators. **Wave 8 (`v1.2.0-wave8` , Track 8C)** scales TARQ RAREBAL-LDLQ **10.02× from 56 to 561 hot experts** (307 MI50 + 217 P150A covering **34/34 pinned L1 shards** + 37 MTP; 0 Dual Receipt violations; mean SwiGLU cosine **0.88357 → 0.95228**, **+4.221 dB** SNR) via a 3× MI50 hardware FP64 blocked Cholesky solver (46.2 s → 1.51 s/expert).
2. **True Pre- `shared_experts` Dual-Stream P150A || MI50 Overlap & 91/91 Zero-Scratch `gfx906` Kernels (Track 8A):** Forking a high-priority `router_stream` onto a 4-slot **25,680 KiB host-mapped `/dev/shm` fast ring** (100,000 PCIe coherency iterations, `stale_reads = 0`) and ordering `ev_p150_armed_ring` **strictly before `shared_experts`** eliminates GPU occupancy queuing on P150A wake-up. Paired with 1-CTA `k_pack_mi50_routed_pairs_compact256` and **91/91 runtime + 3/3 verify8 `gfx906` kernels at `ScratchSize == 0`** , Wave 8 cuts MI50 MoE service time **2.01× (57.98 → 28.88 ms/step)** and reduces non-overlapped P150A wait **−51.9% (74.39 → 35.75 ms/step, saving 18.57 ms/step of parallel overlap)**.
3. **7-Shape MeshTrace (`M = 1..1,024` incl. `M = 16` Tree), 1M Tiered CSA2 KV, & Pure-C++20 Anthropic SSE (Tracks 8B, 8D, 8E):** Verifies an entropy-adaptive `M = 8 ↔ M = 16` **DSpark Kolmogorov–Dirac causal tree** inside P150A's existing 32-row L1 `x_decode` tile (**0 B extra L1**, 247.82 MiB pure active L1 across 2× P150A, 7/7 MeshTrace at `parity_cos = 1.0000000` : 0.73 ms M=1, 2.52 ms M=8, **4.53 ms M=16**, `max_TV = 2.793e-16`), unlocks **1,048,576 (1M) active CSA2 context tokens** via 64-bit KV strides + coherent mapped pinned-DDR4 warm KV (268/270/960 MiB free HBM2, exact 1.00000000 cosine at pos 1,048,575), and serves native `/anthropic/v1/messages` SSE + mid-turn POSIX search in pure C++20.

> **Canonical Wave 8 Production Receipts (`PASS 5/5` Audit · Zero Python · 1M Context · Zero Scratch Spill)**
> Verified across live agentic GUI sessions up to **217,367** (`M = 16` auto tree, peak `τ = 4.62`) and **417,556 active context tokens**, Wave 8 accelerates live `<think>` + 9-tool DSML Stream #1 decode **+17.2% to 37.93 tok/s (`τ = 4.03` , 75.6% acceptance vs. 67.0%)** and 27K–44K multi-tool reasoning **+16.7% to 35.36 tok/s (`τ = 3.95`)**, while preserving **614.8 tok/s** 20L CED fresh prefill (>500 tok/s past 146K ctx), **2.31B–2.68B tok/s** hot LCP, and **220.74 GB/s** warm Engram hydration.

1. Full-Precision-Preserving WHT128-HLUT2 & 561-Expert TARQ RAREBAL-LDLQ

1.1 Heterogeneous Subsystem Precision Split & Failure Modes of Stock 2-Bit Formats

DeepSeek-V4.1-Flash-1017 Subsystem	Params	Format	bpw	Physical Tier & Wave 8 Fidelity Role
Routed SwiGLU [216..383] (168/layer)	**~229.2B**	**WHT128-HLUT2**	**2.125**	**2× TSMC 6nm Blackhole P150A (64 GiB GDDR6 + 247.82 MiB pure L1, 217 RAREBAL incl. 34/34 L1 shards) → HiFi2 MVMUL into 8 FP32 Dst tiles.**
Routed SwiGLU [0..215] + 3×128 MTP	**~294.7B**	**WHT128-HLUT2**	**2.125**	**3× AMD Instinct MI50 (96 GiB HBM2, TP3, 307 main + 37 MTP RAREBAL) → 4-op register FMA + FP32 dot/HostRing.**

DeepSeek-V4.1-Flash-1017 Subsystem	Params	Format	bpw	Physical Tier & Wave 8 Fidelity Role
Embed, LM Head, RMSNorm, mHC, Routers	~2.9B	BF16 / FP32	16–32	3× MI50 HBM2 (TP3): unquantized routing gates, n=4 Sinkhorn mixing, & FP32 residual stream.
40 CSA2 Attn, HSI, Shared Experts, DSpark	~32.8B	Row INT8 + FP32	8.01	3× MI50 HBM2 (TP3): cos > 0.9999 across all 40 layers; **tiered 1,048,576-token CSA2 KV**.
External 1017 Engram Memory (L1 & L14)	~189.0B	Native FP8+FP16	8.25	384 GiB Host DDR4 (188.83 GiB locked, 220.74 GB/s warm pre-fault in 0.92 s): bitwise-exact HF tables.

Candidate 2-Bit Format	Eff. bpw	Exact Failure Mode Across Blackhole P150A (Tensix SRAM) & MI50 (`gfx906`)
Stateful Trellis (QTIP / EXL3)	2.00–2.05	Sequential LCG/bitshift state creates loop-carried dependencies across 32-wide Tensix SFPU lanes, stalling 16-subtile L1 CB unpacking by >6×.
Lattice Codebook (QuIP# E8P)	2.06–2.13	Tensix tile-row `SFPLOAD` lacks arbitrary per-lane L1 scatter/gather for 256-entry 8D tables; triggers 8–16-way LDS bank conflicts on `gfx906`.
Unrotated Hardware `Bfp2_b`	2.50	Channel spikes collapse shared exponents (cos = 0.729); hits TT-Metal LLK bugs (#58712, #56682, #57683); overflows 160 GiB DRAM by +24.6 GiB.
WHT128-HLUT2 (PolarQuant)	**2.125**	**Zero LUT gathers, zero sequential state, zero LLK bugs: orthonormal WHT128 Gaussianizes 128-wide blocks and 2-bit centroids expand via 1 affine FMA into L1 `Bfp8_b` → FP32 `Dst` & `v_dot2_f32_f16`.**

1.2 Mathematical Formulation & Track 8C 561-Expert 3× MI50 FP64 REREBAL-LDLQ Scale-Up

Rotating weights by orthonormal $Q_b = (1/\sqrt{128})\, H_{128}\, diag(s_b)$ preserves inner products while Gaussianizing weights to $N(0, \sigma_{r,g}^2\, I_{128})$ (mirroring Thinking Machines Lab's End-to-End 4-Bit Training block Hadamard outlier suppression). Offloading `B = 128` blocked FP64 Reverse-Cholesky (QuIP# LDLQ + TARQ REREBAL) onto `gfx906`'s 1:2 FP64 units via `rocblas_dgemm` across 3× MI50 cuts repack time **30.6× (46.2 s → 1.51 s/expert)**, scaling verified overlays **10.02× from 56 to 561 hot experts** (4.91 GiB in-place overlay, **0 B extra VRAM**, **34/34 P150A L1 shards**, 0 Dual Receipt violations, mean SwiGLU cos **0.88357 → 0.95228**, **+4.221 dB** SNR):

$$\hat{w}(s, m) = (-1)^s\, \sigma_{r,g}\, (c_0 + m(c_1 - c_0)), \quad \lambda = \mathrm{tr}(H_{common}) / \mathrm{tr}(H_{rare}), \quad H_{bal} = (H_{common} + \lambda\, H_{rare}) / M_{cal} + 0.015\, \bar{d}\, I_K$$

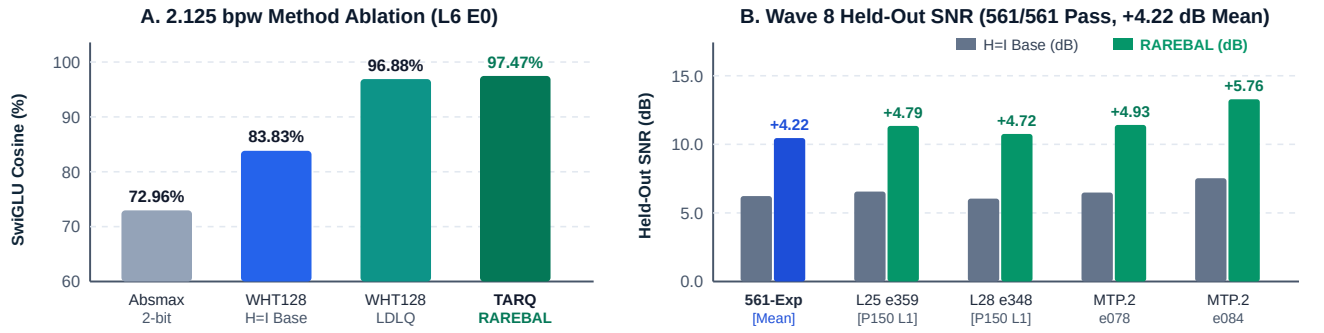


Figure 1: (A) Layer 6 Expert 0 ablation at 2.125 bpw (97.47% SwiGLU cosine, +13.02 dB SNR, −24.7% rare MSE). (B) Wave 8 Track 8C held-out receipts (`M_cal ≥ 11,272`, `M_eval = 512`): 561/561 hot experts (34/34 P150A L1 shards + 307 MI50 + 37 MTP) pass Dual Receipt gates (+4.221 dB mean SNR).

2. Lead Silicon: 2× Custom TSMC 6nm Blackhole P150A, 7-Shape MeshTrace, & Dual-Stream Overlap

Lead MoE Tier: 2× TSMC 6nm Tenstorrent Blackhole P150A (TP2)

4× 800G QSFP-DD Direct card-to-card **3.2 Tbps** **`FABRIC_1D`** mesh absorbs 100% of TP2 `all_reduce` & sync off Card #1's PCIe x4 riser.

7/7 MeshTrace 1.5 GiB trace replays 40 layers at `parity_cos = 1.0000000` : **0.73 ms** M=1, **2.52 ms** M=8, **4.53 ms** M=16 (**0 B extra L1**).

34/34 RAREBAL L1 **17/17 RAREBAL shards/card (152.40 MiB)** + L1 `x_decode & finish_decode[0..1]` = **123.91 MiB/card (247.82 MiB pure L1)**.

Dual-Stream Ring, 91/91 Zero-Scratch MI50, & 1M CSA2 Tier

Pre-Shared Ring 4-slot **25,680 KiB** **`/dev/shm`** ring on `router_stream` fires before `shared_experts` , saving **18.57 ms/step** overlap.

3× MI50 HBM2 **91/91 zero-scratch** runtime kernels + 1-CTA `compact256` (**2.01× faster MoE**: 28.88 ms/step) + SGPR M=8↔16 Dirac tree.

1M CSA2 + SSE **1,048,576-tok** tiered HBM2 + mapped-DDR4 CSA2 KV, 220.7 GB/s Engram, & pure-C++20 `/anthropic/v1/messages` SSE.

Figure 2: Wave 8 heterogeneous architecture pairing 2× custom TSMC 6nm Blackhole P150A ASICs (34/34 RAREBAL L1 shards, 7/7 MeshTrace) with pre-`shared_experts` dual-stream IPC, 91/91 zero-scratch `gfx906` kernels, and 1M-token tiered CSA2 sparse attention.

2.1 7/7 MeshTrace Verification & Pre-`shared_experts` Dual-Stream Doorbell (Tracks 8A & 8B)

Isolating `FABRIC_1D` CCL semaphores in `kP150L1SmallSizeBytes = 2,048 B` leaves `1,459,200 B/bank` across 110 Tensix cores/card. Because `x_decode` (`[1,1,32,5120]` BF16 = `4,096 B/bank`) is already a 32-row tile, adding `M = 16` causal-tree verification requires **0 B extra L1** across all 7 `P150_SELFCHECK` shapes (`M=1` : **0.73 ms**; `M=6` : **2.22 ms**; `M=8` : **2.52 ms**; `M=16` : **4.53 ms**; `M=64` : **11.35 ms**; `M=512` : **14.74 ms**; `M=1,024` : **30.24 ms**). On `tp = 0` , high-priority `router_stream` writes `fast_slots[rslot]` + system-scope release doorbell (100,000 PCIe iters in 3,978.30 ms, `stale_reads = 0`) and records `ev_p150_armed_ring` **strictly before** **`shared_experts`** :

Blackhole P150A & Dual-Stream Invariant (Wave 8)	Exact Byte / Cycle Arithmetic	Verified Hardware Receipt
Usable L1 per Core & Tier-1b Bank Usage (17 Shards/Card)	`1,453,184 B` (incl. 256 KiB CB guard) ≤ `1,459,200 B` ceiling	**+8,832 B Post-Trace Margin**
Pure On-Die Active L1 + 34/34 RAREBAL Shards (`0 B` Extra @ M=16)	**`50,004,608` (`ws1`) + `20,480` (LUT) + `79,902,720` (17/17 RAREBAL)**	**129,927,808 B (247.82 MiB 2×)**
4-Slot Fast Ring + Zero-SDMA Host-Mapped `/dev/shm` Mailbox	`4 × 1,314,816 B` fast slots + `21,037,248 B` base mailbox	**26,296,512 B (25,680.19 KiB)**
1-CTA `compact256` MI50 MoE & Parallel P150A || MI50 Overlap	**`p150 = 54.33 ms – ovlp = 18.57 ms = p150w = 35.75 ms/step`**	**2.01× MI50 MoE · –51.9% Wait**

3. Algorithmic & Protocol Stack: M = 8 ↔ 16 Tree, 1M CSA2 KV, & Pure-C++20 Anthropic SSE

- **Entropy-Adaptive M = 8 ↔ 16 Kolmogorov–Dirac Causal Tree (Track 8B):** Extends SGPR-packed M=8 (`max_TV = 1.735e-16`) to 16-node SGPR `TreeAncMask256` attention (`--dspark-tree-width auto|8|16` , `ScratchSize == 0` , analytical **`max_TV = 2.793e-16 < 1e-15`** , `514.3 ns/step` across 1M sweeps), retaining M=8 (2.52 ms/layer) as throughput default while verifying live `auto` /M=16 across 39 GUI turns up to **217,367 active tokens** (peak `τ = 4.62`).
- **91/91 Zero-Scratch `gfx906` Kernels & 1,048,576-Token (1M) Tiered CSA2 Context (Track 8D):** All **91 runtime + 3 verify8** **`gfx906`** **kernels** compile at `ScratchSize == 0` , 0 device `v_sin_f32` / `v_cos_f32` , and 0 BAR reads. 64-bit `size_t` KV strides + 262,144 hot HBM2 `r=2` slots (524,288 hot tokens) + coherent mapped pinned-DDR4 warm KV unlock **`max_seq_len = 1,048,576`** with **268 / 270 / 960 MiB free HBM2** and exact **1.00000000** cosine at pos 1,048,575.
- **Pure-C++20 `/anthropic/v1/messages` SSE & Mid-Turn POSIX Search (Track 8E):** Native zero-Python Anthropic Messages translation, CORS preflight, `thinking_delta` / `tool_use` SSE streaming, and optional `--search-url` mid-turn POSIX HTTP search with 100% `RadixBlockCache` prefix parity.

4. Wave 8 Production Receipts (`PASS 5/5`) & Frontier MoE Comparison

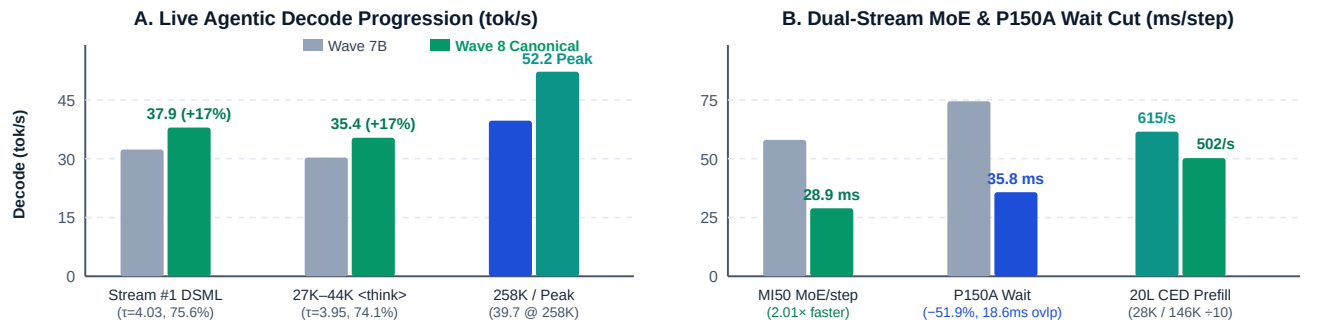


Figure 3: (A) Matched Wave 7B vs. Wave 8 live GUI decode (+17.2% Stream #1 to 37.93 tok/s, +16.7% multi-tool to 35.36 tok/s). (B) 2.01× faster MI50 MoE (28.88 ms/step) and −51.9% non-overlapped P150A wait (35.75 ms/step, 18.57 ms/step parallel overlap saved).

Live Metric / Stage	Wave 7B (`v1.1.0-wave7b`)	Wave 8 Canonical (`v1.2.0-wave8`)	Verified Delta / Physical Floor
Stream #1 Decode (~9.2K `<think>` + 9 DSML Tools)	**32.37 tok/s (τ=3.56, 67.0%)**	**37.93 tok/s (τ=4.03, 75.6%)**	**+17.2% tok/s · +8.6% DSpark acc**
Multi-Tool `<think>` + DSML (26.6K–44K ctx)	30.30 tok/s (τ=3.45, 64.6%)	**35.36 tok/s (τ=3.95, 74.1%)**	**+16.7% tok/s (M=16 peak τ=4.62 @ 217K)**
MI50 MoE Service Time & Non-Overlapped P150A Wait	**57.98 ms MoE · 74.39 ms wait**	**28.88 ms MoE · 35.75 ms wait**	**2.01× MoE · −51.9% wait (18.57 ms ovlp)**
Calibrated TARQ RAREBAL-LDLQ Hot Experts	56 (502.8 MiB, 30/34 L1)	**561 (10.02×, 34/34 P150A L1)**	**Mean SwiGLU cos 0.88357 → 0.95228 (+4.22 dB)**
Active CSA2 Context Ceiling & Free MI50 HBM2	**524,288 (417.6K live)**	**1,048,576 (1M) (217.4K & 417.6K live)**	**268 / 270 / 960 MiB free HBM2 (0 faults)**
20L CED Fresh Prefill & Peak Hot LCP	614.8 tok/s · 2.68B tok/s	**614.8 @ 28K · >500 @ 146K**	**2.31B–2.68B LCP · 220.74 GB/s Engram**

Engineering Wave	Key 2× TSMC 6nm Blackhole P150A, Quantization, & Runtime Breakthroughs	Prefix / Context	Live Decode
Waves 1–5	WHT128-HLUT2 (2.125 bpw → FP32 Dst), 20L CED, zero-BAR HostRing, 22 L1 shards (98.6 MiB), 1.5 GiB MeshTrace	≥2,000 cold · 512K	9.4 → 47.5–52.2 peak
Wave 6 / 6C (`v1.0.0`)	Zero-trig Givens RoPE, FP32 Ring+mHC, 56/56 TARQ RAREBAL-LDLQ overlay, Barrett Engram, ThinkLoopGuard rewind	1.998M/s switch (72 ms)	37.1 (10K think) · 39.7 @ 258K
Wave 7B (`v1.1.0-wave7b`)	34 TP2 L1 shards + L1 `x_decode` (247.82 MiB pure L1), zero-SDMA shm, 88/88 zero-scratch `down32` , SGPR M=8 tree	614.8 @ 28K · 417.6K live	30.3 @ 57K · 36.84 (+38%)
Wave 8 (`v1.2.0-wave8`)	**561-expert GPU FP64 RAREBAL (34/34 L1), pre- `shared_experts` 4-slot ring + 1-CTA `compact256` , 7/7 MeshTrace (M=16), 1M CSA2, Anthropic SSE**	**1,048,576 (1M) · 2.68B LCP**	**37.93 (+17.2%) · 35.36 (+16.7%)**

Open-Weight Frontier MoE	DeepSeek-V4.1-Flash-1017 (Wave 8)	Inkling-2-Small / Tess-4.5-594B	Mistral Large 4 (1.05T / A52B)	Heterogeneous 160 GiB Workstation Verdict
Total / Active Params	**748.6B / 8–16B (+189B DDR4 Engram)**	594B MoE / 17B active (1.8%)	1.05T MoE / 49–52B active (top-4)	DDR4 Engram preserves 160 GiB accelerator DRAM for compute.
Weights & 1M KV	**158.5 GiB @ 2.125b · 890 MiB CSA2 @ 1M**	**~165 GiB @ 2.1b · ~32 GiB KV @ 1M**	**610 GB NVFP4 (~290 GiB @ 2.1b)**	**Only DS-V4.1-Flash fits 160 GiB DRAM at 1M context (0 CPU MoE).**
DeepSWE / Speed	**77.0% DeepSWE · 35.4–37.9 tok/s**	72.1% DeepSWE / 494.3 AUC	61.7% DeepSWE / 478.4 AUC	Highest DeepSWE (77.0%) + 3–5× faster local DSpark tree decode.

Credits: bespokeontology (Hardware & Systems Direction) · Google Antigravity & Gemini (Google DeepMind — C++/TT-Metal/HIP Co-Engineering, Dual-Stream, 561-RAREBAL & 1M CSA2) · Self-hosted DeepSeek-V4.1-Flash-1017 (PASS 5/5 Audits)

v1.2.0-wave8